

How Is the Systems Engineering Viewpoint Different from My Current Working View?

My Current Working Viewpoint

My current working view is a hybrid of two roles, and neither is clinical. As team leader on a biomedical engineering design team building a minimally invasive bladder stone removal device, I think in terms of an artifact: geometry, material selection, force transmission, bench performance, and whether the prototype can be carried through patent filing toward a commercial path. As a strategy and M&A analyst — at Stanford Medicine, at the NATO Centre of Excellence for Military Medicine, and at Churchill Capital — I think in terms of a decision: gather evidence, run a root cause analysis, and converge on a recommendation a stakeholder can act on inside a defined engagement window. Both lenses are productive, and both are narrower than they feel from the inside. The device lens optimizes a part. The analyst lens optimizes an answer. Systems engineering optimizes neither. It optimizes the capability that emerges when parts and answers are assembled into something that must function in a real environment, over a long life, for people whose incentives do not align.

What Counts as “the System”

Kossiakoff's definition — a set of interrelated components working together toward some common objective — reframes my device work immediately. The bladder stone device is not the system. It is one component of a system that also includes the urologist's technique, the cystoscope it must interface with, the operating room's sterilization and reprocessing loop, the training a first-time user receives, the disposal pathway for single-use elements, the reimbursement code that determines whether a hospital will buy it, and the patient's post-operative course. My team has been optimizing the black box while treating everything outside it as someone else's problem. The context diagram introduced in this module enforces the opposite discipline: define the boundary first, enumerate everything that crosses it — inputs, outputs, infrastructure, personnel, laws and regulations — and only then design inside. Those externalities do not disappear because we deferred them. They arrive late, when accommodating them costs the most.

Needs First, Not Solutions First

The sharpest difference is sequencing. Montoya's warning that healthcare's characteristic failure is to meet a problem and immediately buy the device or the widget describes much of what analytical work rewards. My NATO engagement asked me to identify and prioritize five AI-enabled solutions for forward medical support operations. The deliverable was denominated in solution units from the outset. The systems engineering method inverts that order: identify operational needs in contested environments, translate those needs into quantified requirements, decompose the requirements into the activities a capability must perform, and only then allocate physical solutions — hardware, software, or a person in the loop — against those activities. “Deploy AI-assisted triage” is a solution wearing the costume of a need. “Sort N casualties by survivability within M minutes, at a stated accuracy, without reliable communications reachback” is a requirement, and a requirement can be verified. The second formulation would have made my recommendations testable rather than merely persuasive.

How Long the System Lives

My working view is bounded by engagements: a ten-week externship, a semester design cycle, the interval between deal announcement and close. The systems engineering lifecycle is bounded instead by the system's existence. INCOSE's six stages — concept, development, production, utilization, support, and retirement — place most of a system's life after the point at which my current work would already have declared victory. For the bladder stone device, “prototype to market” is my present end state; in lifecycle terms, market entry is roughly where utilization begins. The questions that actually determine whether the device helps anyone — integrated logistics support, maintenance, obsolescence of a supplier's component, and eventual retirement and disposal — sit entirely downstream of my horizon. Kossiakoff makes the same argument in fewer words: post-development is a stage, not an afterthought.

Whose Viewpoint Counts

I have handled stakeholder plurality intuitively — iterating with NATO representatives through feedback sessions, presenting cross-functionally to Stanford Medicine stakeholders — but intuition is not method. The blind-men-and-the-elephant image from this module makes an epistemic claim, not a claim about manners: the surgeon, the sterile processing technician, the biomedical equipment technician, the hospital's value analysis committee, and the patient each describe a real part of the same object, and none of them is wrong. Senge's "I am my position" explains why the fragmentation persists, since each stakeholder's incentives are local and rational. Pronovost makes the same observation at industry scale when he argues that American health systems have bought or built the parts without aligning them around a goal — precisely a system failing the definition of a system. My current view treats conflicting stakeholder input as noise to be reconciled near the end of a project. Systems engineering treats it as the primary source of requirements, elicited at the beginning.

What "Done" Means

A consulting engagement ends when the deck is delivered; a design sprint ends when the prototype performs on the bench. Systems engineering accepts neither as closure. The method terminates in verification and validation — did we build the thing right, and did we build the right thing — with traceability running from each verified result back to the originating need. The INCOSE handbook argues that this traceability is much of the value systems engineering delivers in government and defense contexts, which bears directly on the NATO work: a recommendation carrying an audit trail from operational need to requirement to test outcome survives scrutiny in a way that a well-argued slide does not. The same contrast governs how I handle failure. My use of root cause analysis is retrospective — something broke, so I diagnose it. Systems engineering risk management is prospective and runs the length of the lifecycle, asking what could fail and what the mitigation is before anything has.

What I Intend to Change

Three shifts follow. First, I will build a context diagram for the bladder stone device before the next design review and treat every flow crossing the boundary as a source of requirements rather than as background environment. Second, I will stop accepting solution-shaped problem statements; when a stakeholder hands me one, I will work backward to the underlying need and write the requirement any candidate solution would have to satisfy. Third, I will extend my horizon past the deliverable, so that for any recommendation I can name who supports it, who maintains it, and how it ends. Senge supplies the constraint on all three: none of this survives contact with an organization unless it can be shared. A systems view held only in my own head is a private mental model, not a capability. Making that model explicit and legible to the clinicians, technicians, and executives who each see a different part of the elephant is what converts the viewpoint from a way of thinking into work an organization will adopt.

References

- INCOSE. (2023). Systems engineering handbook: A guide for system life cycle processes and activities (5th ed.). INCOSE-TP-2003-002-05.
- Kossiakoff, A., Biemer, S. M., Seymour, S. J., & Flanigan, D. A. (2020). Systems engineering principles and practice (3rd ed.). Wiley.
- Montoya, M. (n.d.). Module 1: Framework, context, and motivation [Course lecture]. 655.662 Introduction to Healthcare Systems Engineering, Johns Hopkins University.
- Pronovost, P. (2019). New narratives for health care [Address]. The City Club of Cleveland.
- Senge, P. M. (2006). The fifth discipline: The art and practice of the learning organization. Doubleday/Currency.